

BELLINGHAM, WA, USA

WMO: 727976

Lat: 48.794N

Lon: 122.537W

Elev: 149

StdP: 14.62

Time Zone: -8.00 (NAP)

Period: 94-19

WBAN: 24217

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	20.5	24.7	2.6	6.3	25.7	9.7	9.1	28.4	32.6	43.7	28.8	43.5	11.9	20	0.558

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	16.3	80.2	65.4	76.8	63.9	73.6	62.2	66.5	77.8	64.8	75.0	63.2	72.2	6.4	280

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
61.7	82.6	70.7	60.6	79.3	69.0	59.1	75.3	67.1	31.1	78.1	29.8	75.1	28.6	72.2	74.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
25.4	20.7	18.1	DB	15.5	86.9	4.7	3.9	12.1	89.7	9.4	91.9	6.7	94.1	3.3	97.0
			WB	13.4	69.9	4.6	1.9	10.1	71.2	7.4	72.3	4.8	73.4	1.5	74.8

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	50.8	40.1	41.1	44.7	49.0	54.7	59.2	63.5	63.5	58.4	50.6	44.2	39.3	(6)
	DBStd	10.05	7.05	5.79	4.99	4.34	4.53	4.03	3.94	3.59	4.12	4.74	6.59	6.85	(7)
	HDD50	1389	310	251	174	67	11	0	0	0	1	49	191	335	(8)
	HDD65	5265	771	668	629	479	321	181	77	72	201	445	624	797	(9)
	CDD50	1661	3	2	10	38	155	276	417	418	253	69	17	3	(10)
	CDD65	65	0	0	0	0	0	7	29	26	3	0	0	0	(11)
	CDH74	630	0	0	0	2	20	94	248	219	47	0	0	0	(12)
	CDH80	96	0	0	0	0	1	12	48	30	5	0	0	0	(13)
(14) Wind	WSAvg	7.2	8.1	7.8	8.1	6.9	6.7	7.3	7.3	6.6	5.3	6.0	7.9	8.1	(14)
(15) Precipitation	PrecAvg	35.8	4.6	3.4	3.0	2.8	2.4	1.8	1.3	1.5	1.8	3.3	5.0	4.7	(15)
	PrecMax	47.2	10.6	8.7	5.4	6.1	5.9	4.8	2.9	4.8	4.7	7.9	11.6	10.0	(16)
	PrecMin	25.4	0.8	0.5	0.7	1.3	0.4	0.2	0.0	0.0	0.0	0.3	1.4	1.0	(17)
	PrecStd	5.5	2.5	1.9	1.1	0.9	1.2	1.0	0.9	1.2	1.2	1.7	2.3	1.9	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	57.8	58.4	64.0	70.9	76.7	81.8	85.1	83.7	78.9	68.4	62.6	57.8	(19)
		MCWB	51.8	49.5	52.3	57.1	61.1	64.5	67.7	66.5	63.1	56.9	54.6	51.8	(20)
	2%	DB	54.3	54.5	58.9	64.4	71.6	76.5	80.2	79.4	74.4	64.3	57.8	54.0	(21)
		MCWB	49.6	47.4	49.9	53.4	58.8	63.4	65.9	65.4	61.9	55.7	53.0	49.0	(22)
	5%	DB	52.1	52.0	55.5	60.7	67.4	71.8	76.3	76.0	71.0	61.2	55.4	50.8	(23)
		MCWB	48.0	46.2	48.7	51.7	57.4	60.7	64.1	63.9	60.9	54.5	51.3	46.7	(24)
	10%	DB	49.8	49.8	53.0	57.2	63.7	67.9	72.9	72.8	67.4	58.6	53.4	48.3	(25)
		MCWB	46.2	45.0	47.0	49.5	55.7	58.7	62.2	62.4	59.2	53.2	49.8	44.8	(26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	53.2	52.1	53.7	58.8	62.5	66.7	69.7	68.4	65.7	60.5	56.5	53.2	(27)
		MCDB	55.9	55.9	60.4	68.5	73.0	79.1	81.9	80.2	75.2	65.3	60.0	56.7	(28)
	2%	WB	50.6	49.2	51.4	55.0	60.1	63.9	66.8	66.3	63.5	57.6	54.0	49.8	(29)
		MCDB	53.4	52.5	56.9	62.4	69.4	74.8	78.4	77.5	72.1	61.5	57.0	53.2	(30)
	5%	WB	48.4	47.2	49.6	52.6	58.2	61.4	64.8	64.6	61.7	55.6	52.0	47.0	(31)
		MCDB	51.3	50.7	54.6	58.5	66.0	70.4	74.9	74.7	69.1	59.8	55.0	50.0	(32)
	10%	WB	46.3	45.8	47.8	50.8	56.4	59.4	63.0	63.0	60.0	53.9	49.9	45.4	(33)
		MCDB	49.0	49.0	52.4	56.0	63.0	66.8	71.8	71.6	65.9	57.5	53.0	48.0	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	11.0	12.8	13.5	14.7	15.2	14.6	16.3	16.8	16.8	14.1	11.8	10.2	(35)
		MCDBR	13.0	16.2	18.6	21.9	23.5	23.4	23.9	24.0	23.9	18.2	13.0	11.4	(36)
		MCWBR	8.9	10.4	11.1	12.7	12.9	12.0	11.5	11.8	13.0	11.2	8.5	8.2	(37)
	5% WB	MCDBR	11.7	13.1	16.0	18.9	20.4	21.2	22.4	22.3	20.8	15.4	11.6	10.5	(38)
		MCWBR	8.6	9.2	10.2	11.5	11.6	11.4	11.1	11.4	11.8	10.4	8.3	8.3	(39)
(40) Clear-Sky Solar Irradiance	taub		0.314	0.310	0.327	0.353	0.376	0.354	0.349	0.354	0.338	0.332	0.317	0.308	(40)
	taud		2.517	2.554	2.519	2.433	2.381	2.456	2.496	2.493	2.531	2.539	2.523	2.493	(41)
	Ebn at Noon		240	272	283	285	281	287	287	281	276	258	237	226	(42)
	Edh at Noon		19	23	28	33	36	34	32	31	27	23	19	17	(43)
(44) All-Sky Solar Radiation	RadAvg		285	562	865	1325	1711	1857	2046	1729	1217	669	339	228	(44)
	RadStd		41	79	108	131	140	193	158	121	103	81	56	32	(45)

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only	N/A	N/A	N/A	+1.21	N/A	N/A	N/A	N/A	N/A	+23
Regional (72 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page